Guess, Check, Correct

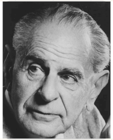
Weekend 1: Vapnik's statistical learning theory

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich

Friday 4 and Saturday 5 September 2026

Popper, 1934: the tetradic schema



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We do not know: we can only guess.

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Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972).

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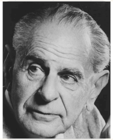

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$\boxed{P_1}$

Problem

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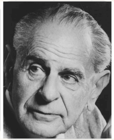

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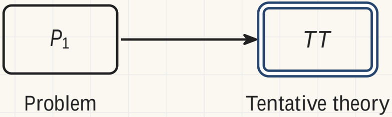

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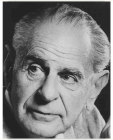

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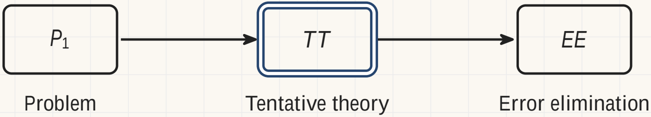

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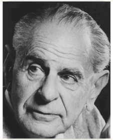

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P_1 → TT → EE → P_2

Problem → Tentative theory → Error elimination → A better problem

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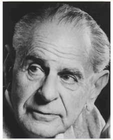

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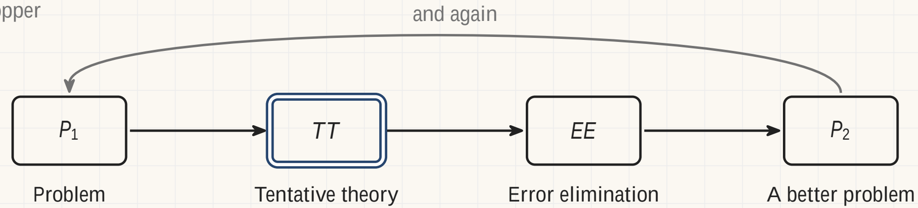

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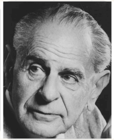

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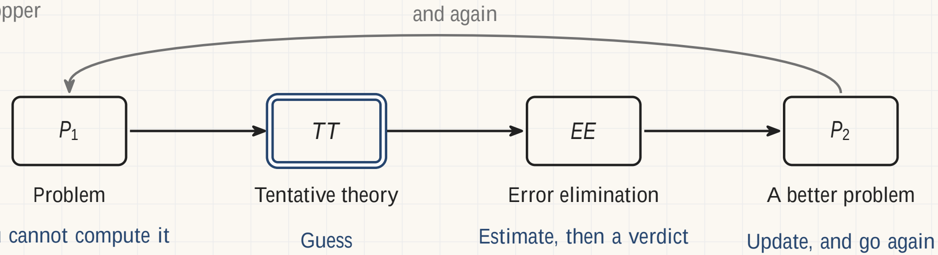

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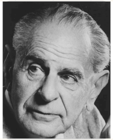

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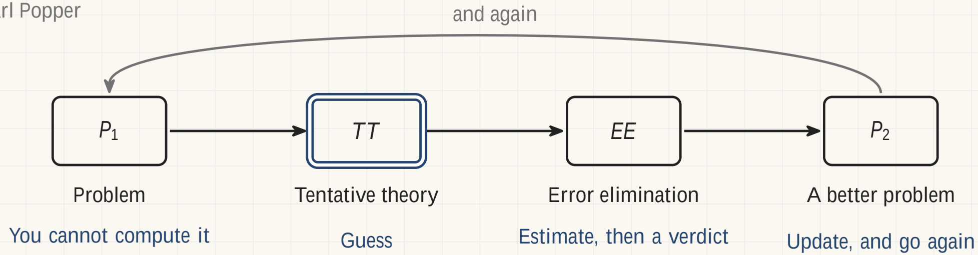

This weekend is that diagram, and how it produces the AI applications you already use.

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Astronomy: the recovery of Ceres, 1801

Carl Friedrich Gauss, 1840.

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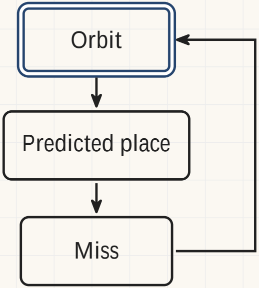

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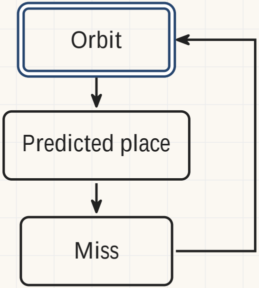
- **December 1801.** Franz Xaver von Zach finds it again.



Computer science: Turing, 1950

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Elliott and Fry

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The loop was written down as the plan for machine intelligence before there was a machine to run it on.

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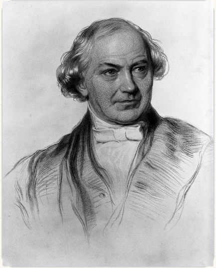
Whewell, 1858: trying wrong guesses



William Whewell, stipple engraving.

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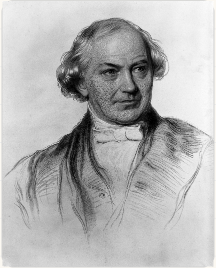

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Wellcome Collection, CC BY 4.0

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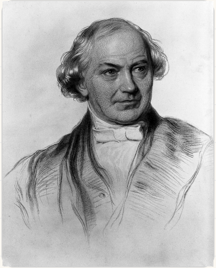

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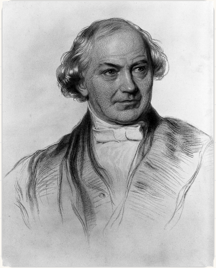

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Most guesses are wrong, and a step that aims too far is how the right one gets found.

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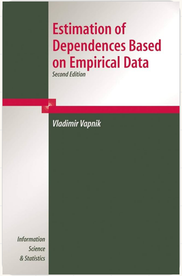
The loop has a name, and the name is a book



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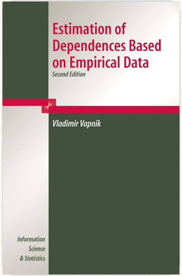

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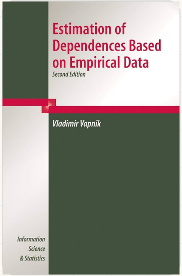

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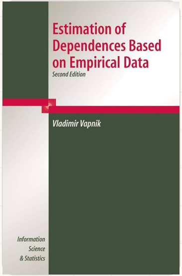

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The book sets out empirical risk minimisation, which is the strategy underneath most of what the rest of this course will build.

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